THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 8: **Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of pairs of chromosomes in humans is: 1

(a) 46 **(b)** 23 **(c)** 22 **(d)** 44

2. A cross between Tt and Tt gives the proportion of tall plants as: 1

(a) 50% **(b)** 75% **(c)** 100% **(d)** 25%

3. A pure-breeding tall plant has the genotype: 1

(a) tt **(b)** TT **(c)** T **(d)** Tt

4. A student states: “Recessive traits disappear permanently in the F1 generation.” The correct response is: 1

(a) Correct **(b)** Correct for colour only

(c) Incorrect — they reappear in F2 **(d)** Cannot be determined

5. For a heterozygous plant, Student A writes Tt; Student B writes TT. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The chance of having a male child is: 1

(a) 75% **(b)** 25% **(c)** 100% **(d)** 50%

7. In Mendel's experiment, green seed colour was: 1

(a) recessive **(b)** co-dominant **(c)** dominant **(d)** absent

8. Genes are segments of: 1

(a) proteins **(b)** RNA only **(c)** lipids **(d)** DNA

9. The phenotype of RrYy is: 1

(a) round green **(b)** wrinkled green **(c)** wrinkled yellow **(d)** round yellow

10. In the F2 of a dihybrid cross, the number of plants with both recessive traits out of 16 is: 1

(a) 0 **(b)** 9 **(c)** 3 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What are the two copies of a gene in a pair called? What happens to them during gamete formation? 2

12. Define genotype and phenotype. 2

13. What is meant by a monohybrid cross? 2

14. A student says sex chromosomes in females are dissimilar. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A plant with violet flowers (VV) is crossed with a white-flowered plant (vv). What will be the F1 and F2 results? Show the cross. 3

16. What are autosomes and sex chromosomes? How many of each do humans have? 3

17. Explain how traits are inherited independently, with a dihybrid example. 3

OR

Why are traits acquired during the lifetime not passed on to the next generation? 3

18. What is the importance of variation in a population? (three) 3

OR

How does Mendel's experiment show that traits may be dominant or recessive? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) A pea plant with round green seeds (RRyy) is crossed with wrinkled yellow seeds (rrYY). Write the F1 genotype and phenotype. 5
(ii) What gametes will F1 produce?
(iii) What proportion of F2 will be round yellow?

OR

(i) What are Mendel's laws of inheritance? 5
(ii) Explain with examples.
(iii) Why are his experiments called scientific?

20. **Case study:** Seema observed that her pea plants with purple flowers produced some white-flowered plants on self-pollination. 5
(i) What is the genotype of the parent?
(ii) Show the cross.
(iii) What fraction is white?
(iv) Which trait is dominant?

OR

(i) How is sex determined in humans? 5
(ii) Draw a flow chart.
(iii) How is sex determined in snails and some reptiles?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND